

Show Your Project — API Specification

Backend: Gin — Golang | Database: PostgreSQL | Frontend: Next.js | Authentication: JWT | Base URL: /api/v1

SUMMARY

API MVP harus mengikuti prinsip: Simple, predictable, secure, and transaction-safe.

TECHNICAL SKILLS

REST API

Gin

Golang

PostgreSQL

Next.js

JWT

Authentication

Authorization

Rate limiting

Input validation

Password hashing

Webhook signature verification

SQL parameterization

Idempotency

Pagination

Search

Filtering

Sorting

Middleware

Layer Architecture

Handler

Service

Repository

Database transaction

Atomic transaction

File upload

Admin API

API OVERVIEW

Backend Show Your Project menyediakan REST API untuk: Authentication, Users, Projects, Categories, Leaderboard, Point Wallet, Point Packages, Payments, Boosts, Admin. Core business flow: User → Create Project → Publish → Buy Point → Boost → Active Boost → Leaderboard.

HTTP METHODS

GET (Retrieve data), POST (Create data/action), PUT (Replace/update data), PATCH (Partial update), DELETE (Delete/archive data).

AUTHENTICATION ENDPOINTS

POST /auth/register, POST /auth/login, GET /auth/me, POST /auth/logout.

USER ENDPOINTS

GET /users/:username, PATCH /users/me, GET /users/me/projects.

PROJECT ENDPOINTS

GET /projects, GET /projects/:slug, POST /projects, PATCH /projects/:id, POST /projects/:id/publish, POST /projects/:id/archive.

CATEGORY ENDPOINTS

GET /categories, GET /categories/:slug, GET /categories/:slug/projects.

LEADERBOARD

GET /leaderboard. Ranking rule MVP: ORDER BY active_boost DESC. Tie breaker: active_boost DESC, earliest boost activation ASC, project_id ASC.

WALLET & POINT PACKAGES

GET /wallet, GET /wallet/transactions, GET /point-packages.

PAYMENTS

POST /payments, GET /payments/:id, POST /payments/webhook. Webhook harus idempotent.

BOOSTS

POST /projects/:id/boosts, GET /projects/:id/boosts, GET /projects/:id/boosts/summary. Validation: points $\geq$ 100; duration 24/48/168 hours.

ADMIN API

Semua endpoint /admin/* membutuhkan JWT + role = ADMIN. Includes dashboard, users, projects, categories, point packages management.

RESPONSE & STATUS STANDARDS

Success format: { success: true, data: { } }. Error format: { success: false, error: { code, message } }. Status convention includes 200, 201, 204, 400, 401, 403, 404, 409, 422, 429, 500.

ARCHITECTURE

Recommended backend structure with cmd/server, internal (config, database, middleware, handler, service, repository, model, dto, router), migrations, pkg (auth, response, validator). Layer: HTTP → Handler → Service → Repository → PostgreSQL.

SECURITY & RELIABILITY

Security requirements: input validation, authentication, authorization, ownership validation, rate limiting, password hashing, JWT validation, webhook signature verification, file validation, SQL parameterization. Financial endpoints should support idempotency.

API PRIORITY & SCOPE

P0: core auth, projects, categories, leaderboard, wallet, point-packages, payments, boosts summary. P1: user profile, project archive, project management, admin APIs, upload API, boost history. Out of MVP: Like, Comment, Follow, Notification, Analytics, Referral, Badge, Social Login.